

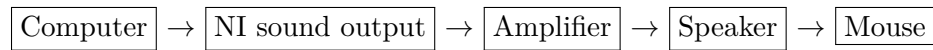
Our sound setup

For mice, with later sheep use in mind

October 8, 2026

We want to play a tone, record the brain's response with our lab's probes, and save when the tone happened. We will build the sound setup for mice and aim to reuse it for sheep.

How the parts fit together



- The computer prepares the tones. The NI box sends the signal, the amplifier drives the speaker, and the speaker plays the sound.
- A microphone checks the sound where the animal's ear will be.
- A timing signal goes to the lab's brain recorder, so we can match each tone to the recorded response.

1. Decide what to play

- Choose how high or low each tone is, how long it lasts, how loud it is, and the gap between tones.
- Prepare and save the full tone sequence before we play it. Keep it with the recordings so we can identify the tone for each trial.

2. Check what reaches the ear

- Fix the speaker in place and record its distance and angle.
- Put the microphone at the ear position. Check every tone's level and waveform, then adjust the output as needed.
- Keep the calibration and background-sound records with the experiment files.

3. Match the sound to the brain recording

- Have the NI box send a timing pulse when each tone starts, and record those pulses with the brain signals. First check that the lab recorder can accept them.
- The pulse marks the electrical start. Match the microphone recording to those pulses, then check when the sound actually arrives.
- Check that the sound and brain-recording times stay aligned throughout a recording.

We still need to choose the mouse tone settings and whether the mouse is awake or sedated. Rice's sheep plan uses sedation; SONIC studied awake sheep. The appendix shows what SONIC did and what we would change.

What we would get

One proposed sound setup

USD

We would get one of each device below. The calibration, computer, and cable rows are budget allowances. **Est.** means our rough estimate; we still need a supplier price or a final parts list.

Item	Why we need it	Budget
Speaker: TDT MF2-M kit	Plays the tones.	\$1,500 est.
Amplifier: TDT SA1	Drives the speaker.	\$2,000 est.
Power unit: TDT ZB1PS	Powers the SA1.	\$750 est.
Sound output: NI USB-6361	Sends tones and markers.	\$3,030
Sound-check kit: Avisoft 116Hm complete	Records the sound at the ear.	\$8,160
Calibration help/reference	Checks sound levels.	\$1,500 est.
Computer, monitor, and storage	Runs playback and saves data.	\$2,000 est.
Cables, timing connections, and mounts	Connects and holds the parts.	\$1,000 est.
Total for this setup	Rounded	\$20,000

The kits already cover several parts

- The Avisoft kit includes the microphone, preamp, sound recorder, cable, stand, and recording software. We get **one complete kit**.
- The MF2-M speaker kit includes a stand and connection accessories.

What we can reuse from the lab

- We use our own lab's probes. This list covers the sound side; it does not budget for new neural hardware or animal preparation.
- If we can use a suitable lab computer and storage, we can remove that \$2,000 allowance.

A little room in the budget

- Add **\$2,500** for price changes, shipping, and tax. That gives us a working sound budget of **about \$22,500**.
- A behavior camera is optional: allow **about \$1,100 extra** with accessories if we need video. A soundproof box is not required.

One choice still matters: the listed microphone is specified from 2 kHz upward. If we want lower tones, we need a different suitable microphone before buying the kit. The appendix explains this for sheep use too.

Price/package sources, October 8, 2026: TDT speaker kit, SA1, power unit, NI price, Avisoft kit, Avisoft prices. Estimate assumptions and alternative prices are saved in the project notes.

Appendix: why these choices?

The speaker and the two parts it needs

- **Speaker (MF2-M):** we want one through-air speaker we could use for mice and later sheep. TDT lists a 1–80 kHz range, which includes SONIC’s explored 1.4–32 kHz range. We still have to measure our chosen tones at the ear.
- **Amplifier (SA1):** the NI box makes the electrical waveform. SA1 provides the power to drive the speaker. TDT supports this combination, and SONIC also used SA1.
- **Power unit (ZB1PS):** SA1 needs TDT’s zBus power connection. This unit powers and houses it, so it belongs with the amplifier.

What SONIC used

Part	SONIC	Our proposed setup
Speaker	TDT MF1, through-air use	TDT MF2-M, current model
Amplifier	TDT SA1, 0 dB attenuation	TDT SA1
Power unit	Model not named	TDT ZB1PS
Tone output	NI USB-6361, 192 kHz	NI USB-6361
Measurement mic	Not named	Avisoft 116Hm complete kit
Shared clock	TimeMachines TM2500C; PTP and 10 MHz	Depends on our lab recorder

- SONIC used 10 ms sine tones with 2 ms fades at each end, in random balanced blocks without gaps. It prepared the waveform before playback.
- It reported about 80 dB SPL at the sheep-ear distance. We have not chosen our level. Its clock server is not in this shopping list.

What we would change for sheep

- Aim to reuse the speaker, amplifier, power unit, NI, computer, and sound recorder.
- Move the mounts and check every tone again at the sheep-ear position. Redo the calibration and timing checks for that recording system.
- Check the lowest tone before buying: the kit’s microphone starts at 2 kHz, so SONIC’s 1.4 kHz tones need another suitable microphone.
- Choose the tone settings and response windows for sedated sheep. SONIC used awake sheep; the same hardware does not guarantee the same brain responses.

References: Perkins et al. (2025), SONIC: A Benchmarking Paradigm for Brain-Computer Interfaces, bioRxiv v1, Methods 6.1–6.2 (p.11), 6.6 (p.13); MF2, SA1, ZB1PS, microphone range. These are proposed choices; we have not tested this setup or its transfer to sheep.